

SSVI Implied-Volatility Surface Dynamics & Realized-Volatility Forecasting

Evidence from S&P 500 Options (2010–2020)

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Abstract

Using ~ 1.5 million filtered S&P 500 option quotes (2010–2020), we calibrate a daily arbitrage-free SSVI implied-volatility surface and ask whether its structural information improves the modelling and forecasting of its own parameters and of realized volatility (RV), relative to random-walk and HAR benchmarks. The central result is *negative with informational content*: a single asset’s own option surface does not beat a parsimonious HAR for RV forecasting through the COVID-19 window, and the surface behaves as a near-efficiently-priced object at the daily frequency. We exercise the full course toolbox — panel regression with fixed/random effects and hypothesis testing, ARMA/ARMAX, VAR, non-stationarity and cointegration, and point/probabilistic forecasting — and add two original contributions: a deliberately asset-agnostic (VIX-free) RV-forecasting study (Notebook C) and an Avellaneda & Stoikov (2008) surface market-making risk engine with an SSVI-based, ES_{95} -calibrated residual add-on (Notebook D).

1. Introduction and Literature Review

Research aims. The project pursues two connected research questions:

- **RQ1 — Predictive content / market efficiency.** Does the structural information in an arbitrage-free IV surface carry incremental out-of-sample (OOS) predictive content for (a) its own future dynamics and (b) future realized volatility, beyond simple benchmarks (random walk, HAR)? A sharpened, deliberately asset-agnostic version asks whether a single underlying’s *own* option surface can supply the regime information usually drawn from the VIX — specific to the S&P 500 — in a form portable to any asset with a liquid option market.
- **RQ2 — Risk-management application.** Can the Avellaneda & Stoikov (2008) optimal market-making framework, which sets quotes for a single instrument, be extended to the full implied-volatility surface with an SSVI-based residual-risk add-on calibrated to a target tail (Expected-Shortfall) coverage?

The first is the efficient-pricing question; the second turns the surface model into a deployable quoting-and-risk tool.

Economic framework. Under efficient option pricing the no-arbitrage IV surface should be close to a martingale at short horizons: if daily surface increments were systematically predictable, a delta-hedged position would extract near risk-free profit, which a functioning market arbitrages away. The competing hypothesis — from the long-memory and path-dependence literature — is that volatility (realized *and* implied) carries persistent, multi-horizon structure that low-order autoregressive models miss but a long-memory model can exploit.

Literature. The methodological backbone rests on three works. We parametrise the surface with the SSVI model of Gatheral & Jacquier (2014), which guarantees freedom from static (butterfly and calendar) arbitrage. The forecasting backbone is the Heterogeneous Autoregressive (HAR) model of Corsi (2009), which approximates the long memory of realized volatility — documented by Andersen, Bollerslev, Diebold & Labys (2003) — with cascaded daily/weekly/monthly components. The hypothesis that the IV surface is itself *path-dependent*, and hence amenable to HAR-type modelling, is the recent contribution of Andr  s, Boumezoued & Jourdain (2025), which directly motivates applying HAR to the SSVI parameters. For the risk engine we build on Avellaneda & Stoikov (2008). Supporting tools: Johansen (1991) and Engle & Granger (1987) for cointegration, Bollerslev (1986) GARCH for conditional heteroskedasticity, and the Diebold & Mariano (1995) test (small-sample corrected) for formal forecast comparison.

2. Data and Methodology

2.1 Dataset (primary focus)

Source and composition. The raw dataset is a panel of daily **S&P 500 European option** quotes spanning 2010-01-01 → 2020-12-31 (~2,769 trading files; ~3.33 million raw rows), each carrying strike, option type, bid/ask/mid, open interest and time-to-expiry (TTE). Zero-coupon **Treasury rates** (1M, 3M, 6M, 1Y, 2Y, 3Y) and the **VIX** are pulled from FRED; the SPX spot is from Yahoo Finance. The data are processed in three notebooks (00_data_preparation, 01_iv_dataset_construction, 02_ssvi_calibration).

Cleaning pipeline (core technique #1). Quality filters remove illiquid and economically meaningless quotes — mid price > \$0.05, time-to-expiry > 7 days, relative bid-ask spread ≤ 30%, and log-moneyness ∈ [-0.40, 0.30]. The implied **forward** is extracted per maturity via put-call parity; Black-76 **implied volatility** is then computed by vectorised Newton-Raphson. At the calibration stage we apply a further **maturity-dependent delta filter**: far-OTM quotes whose absolute Black-Scholes delta falls below a band tightening from 0.30 at the shortest maturities to 0.05 at the longest ($\Delta_{\min}(T) = 0.05 + 0.25 e^{-3T}$) are discarded, removing illiquid wings and keeping OTM puts and calls balanced before the arbitrage-free fit. The funnel is:

Stage	Count
Raw rows	~3,331,302
After domain/liquidity filters	~2,324,008
Black-76 IV retained	~1,500,012
SSVI calibration success	2,633 / 2,768 days (95.1%)

Derived parameter panel. For each trading day we calibrate the SSVI surface, yielding a daily time series of the five parameters ($\alpha, \beta, \rho, \eta, \gamma$). Calibration succeeds on **95.1%** of days with median RMSE_iv < 0.01 and **zero** calendar-arbitrage violations — this calibrated panel is the input to all downstream time-series analysis (Notebooks B and D). The realized-volatility target (Notebook C) is a daily log-RV proxy built from SPX returns.

2.2 SSVI parametrisation and parameter economics

$$\omega(k, \theta) = \frac{\theta}{2} \left\{ 1 + \rho \phi(\theta) k + \sqrt{(\phi(\theta) k + \rho)^2 + (1 - \rho^2)} \right\}, \quad \phi(\theta) = \frac{\eta \theta^{-\gamma}}{1 + \eta \theta^{1-\gamma}}$$

The five calibrated parameters carry direct economic readings (sample means in parentheses): α = log ATM variance / surface level (-3.45), β = term-structure slope (1.19), ρ = skew / leverage effect (-0.77), η = vol-of-vol / smile width (0.75), γ = term-structure decay rate (0.54).

2.3 Econometric toolbox — coverage of the core techniques

The project applies every core econometric technique of the course to the volatility data; the table maps each to its implementation. Section 3 mirrors this mapping, giving each method a self-contained results subsection (§3.1–§3.5), followed by the two original contributions (§3.6).

#	Core technique	Implementation	Reported in
1	Data cleaning, trend & seasonality	Cleaning funnel (NB 00–02); trend via I(d) classification + differencing; seasonality assessed and justified immaterial	§2.1, §3.1
2	Linear regression + hypothesis testing	Panel IV regression with t/F-tests, RESET (NB A); OLS feature-significance pruning (NB C)	§3.2
3	Fixed & random effects	Pooled OLS → Day FE → Day×Maturity FE → Surface-Cell FE, Hausman test FE-vs-RE (NB A)	§3.2
4	ARMA models	BIC-optimal ARMA + ARMAX (exogenous VIX) on 10 series (NB B)	§3.4
5	Point & probabilistic forecasting	Multi-horizon RV point forecasts + DM (NB C); AR-GARCH 80% prediction intervals (NB B); ES₉₅ coverage (NB D)	§3.5
6	VAR models	VAR(2, BIC) on the 5-parameter system + VECM (NB B)	§3.4
7	Non-stationarity & cointegration	ADF + KPSS, Johansen trace, Engle-Granger , rolling Johansen (NB B)	§3.3

Evaluation metrics. OOS skill is measured by $R_{\text{OOS}}^2 = 1 - \sum_t (y_t - \hat{y}_t)^2 / \sum_t (y_t - y_{t-1})^2$ (relative to the naïve random-walk benchmark, so $R_{\text{OOS}}^2 = 0$ for the naïve model by construction), and pairwise comparison by a small-sample-corrected **Diebold-Mariano** test (Diebold & Mariano 1995; HAC Newey-West variance). Sign convention: **DM > 0** ⇒ **model beats HAR**; **DM < 0** ⇒ **model worse than HAR**.

3. Results

3.1 Data cleaning, trend and seasonality (*core technique #1*)

The cleaning funnel (§2.1) retains $\sim 1.5\text{M}$ economically meaningful IV observations from $\sim 3.3\text{M}$ raw rows and produces 2,633 arbitrage-free daily surfaces. **Trend / non-stationarity** is addressed formally in §3.3 (ADF/KPSS): the parameter *levels* are persistent (near-unit-root), so all autoregressive modelling is performed on the stationary **first differences**, and the slow-moving long-memory component is handled by the HAR aggregation (§3.6). **Seasonality**: daily SSVI parameters and log-RV exhibit no material deterministic calendar (e.g. day-of-week) seasonality at this frequency — the dominant low-frequency feature is long-memory persistence, not a periodic component. Where a standard technique is genuinely unsuitable for the data we justify its omission rather than forcing it; accordingly we treat a seasonal decomposition as inappropriate for this dataset and proceed with the persistence/long-memory framing instead.

3.2 Panel regression, fixed/random effects and hypothesis testing (*core techniques #2, #3*) — Notebook A

We regress Black-76 IV on option structural features (forward moneyness k , TTE, a bid-ask liquidity measure) across **~ 1.49 million** observations over 2,768 trading days, climbing a fixed-effects ladder. The within-transformation (time-demeaning) avoids an explicit $\sim 2,768$ -column dummy matrix (≈ 31 GB in double precision).

Specification	Fixed effects	R^2
Pooled OLS	none	~ 0.34
Day FE	trading day	~ 0.80
Day + Maturity FE	day $\times$ maturity bucket	0.86
Surface-Cell FE	day $\times$ maturity $\times$ moneyness	0.96

Hypothesis testing. Forward moneyness and TTE are the dominant, highly significant structural drivers of the IV cross-section. The **Hausman test** rejects random effects ($p < 0.001$) — fixed effects are the correct specification. The **RESET test** flags mild nonlinearity, which the $\log(\text{IV})$ specification reduces. Gauss-Markov diagnostics reveal heteroskedasticity (expected with 1.5M quotes), so HAC standard errors are used throughout; the Jarque-Bera rejection of normality is practically immaterial at this sample size (verified via QQ plot). *Reading*: the surface's **shape** is almost fully explained by moneyness, maturity and day effects — SSVI supplies the geometry, not, as §3.5 shows, an incremental forecasting signal.

3.3 Non-stationarity and cointegration (*core technique #7*) — Notebook B

Stationarity. Each of the five parameter levels gives a *conflicting* ADF (rejects unit root) vs KPSS (rejects stationarity) verdict $\rightarrow$ classified "uncertain / borderline I(1)"; all five first differences are cleanly I(0). This justifies differencing before ARMA/VAR.

Cointegration (α, η). Testing the variance-level/vol-of-vol pair, the full-sample **Johansen** trace test returns **rank = 2** and **Engle-Granger** gives $p = 0.0002^{***}$. In a *bivariate* system rank = 2 is *full rank* — evidence that both series are individually (near-)stationary, not two I(1) series tied by a single long-run attractor (which would appear as rank = 1); the VECM is correspondingly weak OOS ($\alpha R^2 = +0.004$, $\eta R^2 = -0.018$). A rolling Johansen (500-day windows) confirms the link is **regime-conditional** — rank = 1 in 30.0% of windows, rank = 2 in 67.2%, with the estimated rank tracking the VIX level — i.e. a structural feature of the SSVI parametrisation rather than a stable tradeable spread.

3.4 ARMA, ARMAX and VAR modelling (*core techniques #4, #6*) — Notebook B

Data window. 2,632 obs; train 2,105 / test 527 (chronological split).

ARMA / ARMAX — order selection and residual diagnostics. BIC over ARMA(p, q), $p, q \in \{0, 1, 2, 3\}$, selects low orders throughout (levels: $\alpha \rightarrow (1, 0)$, $\beta/\rho/\gamma \rightarrow (1, 1)$, $\eta \rightarrow (2, 1)$; differences: (1,1) everywhere except $d_\gamma \rightarrow \text{MA}(1)$). Each fitted ARMA is screened for the iid-residual assumption: a **Ljung-Box** test (lag 10) finds no remaining serial correlation for β, η, γ (levels) and for $\Delta\alpha, \Delta\rho, \Delta\eta$ (differences), while $\alpha, \rho, \Delta\beta, \Delta\gamma$ retain mild residual autocorrelation; **ARCH-LM** and **Jarque-Bera** reject homoskedastic and normal residuals

throughout. We therefore report HAC standard errors and the AR-GARCH intervals of §3.5 rather than relying on Gaussian OLS inference.

Levels. The surface is essentially unforecastable: only γ beats the random walk (MSE-ratio **0.838**), α and β sit on it (0.998–1.000), and ρ , η are *worse* (1.03–1.13) — coefficient-estimation noise exceeds any exploitable signal.

Differences. Every model posts MSE-ratios of 0.32–0.53 against the naïve "tomorrow's change = today's change" baseline, but that apparent edge is mostly mechanical: the differenced parameters are negatively autocorrelated (-0.36 to -0.09), so a trivial expanding-mean forecast alone achieves 0.38–0.47, and ARMA, ARMAX and HAR all land within a whisker of it (§3.6). Adding exogenous regressors (ARMAX) changes nothing: despite train-set Granger significance of $\Delta\log(\text{VIX})$ and the SSVI fit-RMSE ($p = 0.0001$ –0.005), ARMAX matches ARMA to the third decimal on every series.

Surface-level evaluation (the binding test). Recombining each model's parameter forecasts through the SSVI formula into the full surface (15 strikes $\times$ 5 maturities) and scoring next-day surface RMSE, *no model improves on the sticky (no-change) surface by more than 0.7%* — ARMA 0.9999 $\times$, VAR 0.9927 $\times$, best-per-parameter 1.0133 $\times$ (worse than doing nothing). The parameter-space gains on the differences mostly predict the reversal of yesterday's calibration noise, which leaves no exploitable imprint on the surface a desk actually quotes — the efficient-surface fingerprint where it economically matters.

VAR(2) + VECM (technique #6). A BIC-optimal VAR(2) on the five-parameter system likewise fails to beat the random walk for four of five parameters; **only** γ attains a marginal OOS gain (MSE-ratio = **0.938**). Cross-parameter dynamics add essentially nothing at $h = 1$, consistent with the ARMA result.

3.5 Point and probabilistic forecasting (*core technique #5*) — Notebook C

Design goal (the first original contribution). This is the project's primary forecasting deliverable, and its design is deliberately asset-agnostic. VIX-type regime information is specific to the S&P 500 (the VIX is built from SPX options), so we ask whether a single underlying's *own* option surface — through SSVI levels, ATM IV, skew, and nonlinear stress interactions — can supply the same regime signal in a form **portable to any asset with a liquid option market**. The (non-portable) VIX-augmented HAR is the benchmark to beat; **F3_ATM** (HAR + $\log \sigma_{\text{ATM}}$, requiring only the asset's own options) is the leading *portable* candidate; the M-series progressively adds nonlinear, option-implied stress features.

Setup. Target $y_t^{(h)} = \frac{1}{h} \sum_{i=1}^h \log \text{RV}_{t+i}$ for $h \in \{1, 5, 20\}$; train $\sim 2,135$ / test ~ 534 (the test window includes the COVID-19 shock, $\text{VIX} \approx 83$). A key step (Section 6a) is collinearity control: uncentered λ -interactions reach **VIF = 3,738**, resolved by centering $\tilde{\lambda}_t = \lambda_t - \bar{\lambda}_{\text{train}}$; OLS significance then prunes the feature set. Single-split OOS performance:

Model	Feat.	h=1 R ² _OOS	h=5 R ² _OOS	DM(h=5)	h=20 R ² _OOS	DM(h=20)
Naive	RW	0.000	0.000	—	0.000	—
HAR	3	0.466	0.846	—	0.857	—
HAR+VIX	4	0.488	0.832	-0.58	0.864	+0.52
F3_ATM	4	0.465	0.840	-2.47**	0.846	-1.87*
HAR_rhoJ	4	0.465	0.846	-0.72	0.857	-0.38
M1	4	0.466	0.842	-2.51**	0.849	-1.93*
M4_smooth	9	0.449	0.819	-3.59*	0.806	-2.78*
M4+int	12	0.451	0.825	-2.82*	0.817	-2.13**

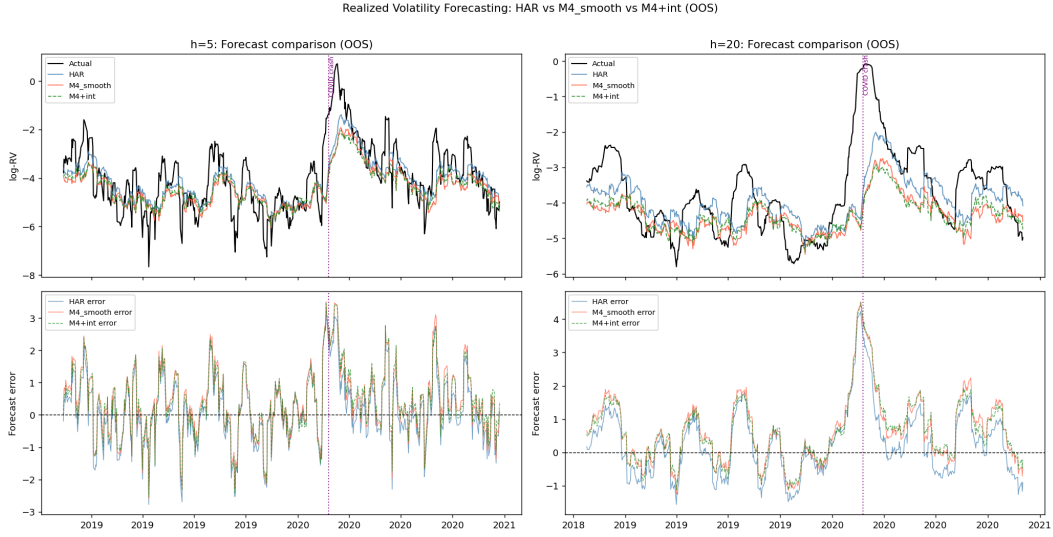


Figure 1: Out-of-sample realized-volatility forecasts versus realized log-RV through the COVID-19 stress window. Top panels: actual log-RV (black) against the HAR benchmark and the option-augmented M4 specifications at horizons $h = 5$ and $h = 20$; the vertical marker locates the March-2020 shock ($VIX \approx 83$). Bottom panels: corresponding forecast errors. The option-implied models track HAR in calm periods but overshoot during the shock, illustrating HAR's parsimony advantage.

Outcome. HAR is the best model: every option-augmented specification is statistically *worse* OOS (DM up to -3.59^{***} at $h=5$), and even the portable F3_ATM trails HAR (-2.47^{**} at $h=5$). In an **expanding-window** re-evaluation (2016–2020 OOS) all option models converge to HAR performance and HAR+VIX is only nominally best (DM not significant, $p \approx 0.26$). The richer models overfit the pre-COVID distribution; HAR's parsimony delivers robustness. The honest, well-identified verdict: a single asset's own option surface carries **no incremental RV-forecasting power beyond HAR** through this stressed window — the portable approach is viable (F3_ATM converges to HAR in calm windows) but not superior on this sample.

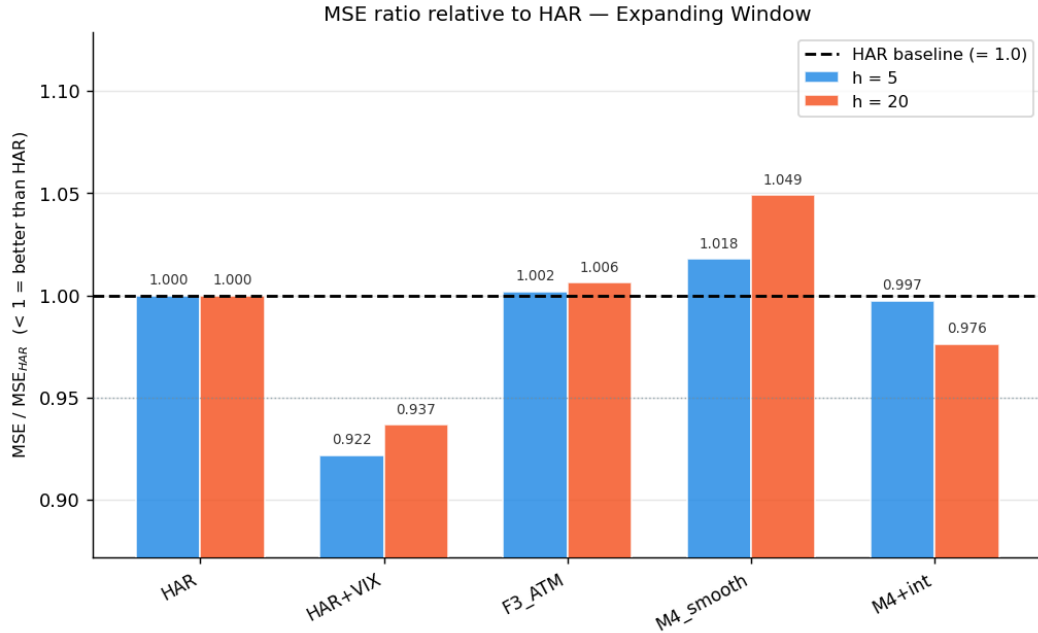


Figure 2: Expanding-window out-of-sample MSE of each forecasting model relative to the HAR benchmark (ratio $< 1 \Rightarrow$ better than HAR), at horizons $h = 5$ and $h = 20$. Under the expanding-window re-evaluation the option-augmented specifications converge to HAR; only the non-portable HAR+VIX falls nominally below unity, and the gap is not statistically significant. The bars complement the single-split R_{OOS}^2 of the table above.

Probabilistic forecasting. Complementing the point forecasts: AR(1)-GARCH(1,1) (Bollerslev 1986) with QMLE errors produces calibrated **80% prediction intervals** for the SSVI parameter paths (Notebook B), and the risk engine below is evaluated by its **ES₉₅** tail coverage (Notebook D).

3.6 Original contributions beyond the core requirements

The two main extras are the *portable RV-forecasting study* of §3.5 (Notebook C) and the *market-making risk engine* below (Notebook D); two shorter findings follow.

(a) A surface market-making risk engine extending Avellaneda-Stoikov — Notebook D

We extend the single-instrument optimal market-making model of **Avellaneda & Stoikov (2008)** to the **full implied-volatility surface** (a 45-point grid, 9 strikes $\times$ 5 maturities, 2,633 days). The Avellaneda-Stoikov logic sets a quoting half-spread around a reservation price that scales with the instrument’s volatility and the maker’s inventory risk; applied surface-wide this gives a vol-scaled baseline spread `spread_AS`. The problem is that this baseline systematically *under-covers* the realised next-day surface move (it ignores the part of surface risk not captured by a simple vol scaling), so we add an **SSVI-based residual-risk add-on** fed by a HAR-J forecast of surface realized volatility, and calibrate it to a target tail (ES_{95}) coverage by maturity bucket.

- **Add-on term structure.** The calibrated add-on coefficient follows $c^*(T) = 4.950 - 3.208\sqrt{T}$ ($R^2 = 0.949$) — strictly decreasing in maturity, economically sensible: short-dated IV moves are more volatile relative to their long-run expectation and require a larger relative buffer. The $\sqrt{T}$ form is consistent with Brownian surface dynamics.
- **Coverage.** The AS baseline alone covers only **57.3%** of next-day surface moves globally (19.4% at 1M, 48.1% at 3M) — far below the 95% target. With the residual-risk add-on (which makes up **74%** of the total spread) coverage reaches **95.9%** globally (97.4% 1M, 96.9% 3M); pointwise surface containment rises from **47.5%** to **82.9%**.
- **Jump robustness.** A bipower-variation jump filter (BPV) flags 29.7% of days — too loose — so a $q_{95}+2\sigma$ threshold (1.4% jump rate) is selected; the HAR-J jump term does not improve the surface-RV forecast (DM $p = 0.290$), so jumps are retained only as a robustness check, not a driver.
- **Sensitivity.** Varying the baseline tightness γ shows a sharp non-linear transition between $\gamma = 1.0$ and $\gamma = 2.0$ (where c^* collapses $\approx 2.2 \rightarrow \approx 0.07$): once the baseline quote is wide enough the vol-scaled AS spread alone nears the 95% target and the add-on becomes redundant — identifying the operating regime in which the SSVI add-on actually adds value.

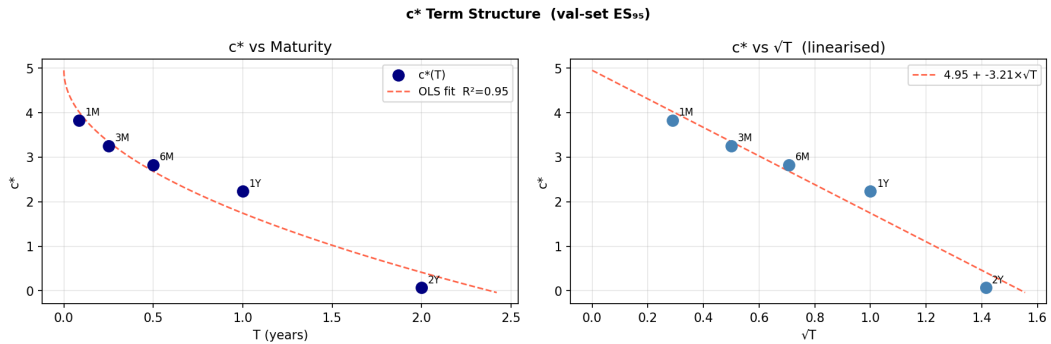


Figure 3: Maturity term structure of the calibrated ES_{95} residual-risk add-on coefficient $c^*(T)$. Left: c^* against maturity T with the fitted decay; right: the same coefficients against $\sqrt{T}$, linearised as $c^*(T) = 4.95 - 3.21\sqrt{T}$ ($R^2 = 0.95$). The monotone decline in maturity is consistent with Brownian surface dynamics — short-dated implied-variance moves are larger relative to their long-run mean and require a proportionally wider buffer.

The engine is **portable** (it uses only the asset’s own calibrated surface) and turns the SSVI model into a deployable quoting-and-risk tool — the project’s most direct operational contribution.

(b) Other findings

HAR on the SSVI parameters: no genuine multi-horizon memory. A HAR(1,5,22) on the *differenced* parameters posts $R^2_{OOS} = 0.51\text{--}0.67$ vs. the random walk for all five series. The differenced parameters carry *negative* lag-1 autocorrelation (−0.09 to −0.36, bid-ask-bounce-like mean reversion), so we benchmark HAR against the **expanding historical mean** (a trivial constant forecast, with a Diebold-Mariano test) and against the ARMA of §3.4 to separate genuine multi-horizon structure from simple shrinkage toward the mean:

Series	HAR R^2_{OOS} (vs RW)	Constant-mean R^2_{OOS} (vs RW)	HAR R^2_{OOS} (vs constant-mean)	DM (p)	ARMA MSE-ratio
d_alpha	0.5664	0.5709	-0.0105	-0.71 (0.48)	0.4272
d_beta	0.5506	0.5534	-0.0064	-0.42 (0.67)	0.4445
d_rho	0.5092	0.5331	-0.0511	-0.80 (0.43)	0.5254
d_eta	0.5642	0.5700	-0.0135	-0.11 (0.91)	0.4471
d_gamma	0.6661	0.6223	+0.1162	+0.64 (0.52)	0.3166

The constant-mean forecast alone reaches $R^2_{\text{OOS}} = 0.53\text{--}0.62$ vs. random walk — essentially matching HAR for $\alpha, \beta, \rho, \eta$, where HAR adds nothing on top of it.

γ is the **partial exception**: HAR beats the constant mean by +0.12 in R^2_{OOS} terms, but the DM test cannot distinguish that edge from noise ($p = 0.52$) and the plain MA(1) of §3.4 (ratio 0.317) matches or beats HAR (0.334) anyway. γ 's predictability is therefore *one-lag mean-reversion*, consistent with its stronger persistence and structural-break profile (S2 Chow tests, rolling-Johansen), not multi-horizon memory.

The Andr  s et al. (2025)-inspired hypothesis — that the parameters' own multi-horizon history forecasts their future innovations — thus finds no support beyond short-memory dynamics, consistent with the efficiently-priced, near-martingale surface documented throughout this section.

PCA / regimes (brief). PCA on surface *changes* gives PC1 = 92.7% (common shift), and a HAR on the PC scores returns negative R^2_{OOS} (-0.004/-0.020/-0.008), consistent with the efficient-surface reading. A 3-state HMM on ΔIV identifies low/mid/high-vol regimes (24%/25%/50% of days); supplementary notebook S2 adds Chow break tests (Volmageddon 2018, COVID 2020), CUSUM stability and Granger-causality checks.

3.7 Limitations and robustness

1. **Distributional shift.** The 2018–2020 test window contains COVID-19, an extreme OOS regime; the negative option-feature results should be re-checked on a calmer window (e.g. 2015–2018).
2. **Single underlying / single SSVI variant.** Results are specific to SPX and to the power-law $\phi(\theta)$; raw SVI, eSSVI or SABR could differ.
3. **VECM / cointegration instability.** Rolling Johansen shows the rank is regime-conditional; the VECM's OOS failure may reflect structural breaks rather than absence of a relationship.
4. **Expanding-window convergence.** Option-augmented models converge to HAR in the expanding window, so part of the single-split deficit is small-sample, not structural.

4. Conclusion

The project's central empirical result is **negative with informational content**: a structurally rich, arbitrage-free SSVI surface does *not* improve on a parsimonious HAR for realized-volatility forecasting, and at the daily frequency the surface behaves like a near-efficiently-priced, near-martingale object (§3.4–§3.6). This answers our first research aim: replacing the SPX-specific VIX with portable, option-implied regime features is *viable but not superior* on this sample — the leading portable model (F3_ATM) converges to HAR in calm windows but cannot beat it through the COVID-19 shock. That HAR wins is itself the economically meaningful finding: backward-looking long-memory structure, not forward-looking surface geometry, carries short-horizon RV predictability here.

The second aim delivers the main operational contribution: extending **Avellaneda & Stoikov (2008)** to the full implied-volatility surface, an SSVI-based residual-risk add-on with a $\sqrt{T}$ -calibrated term structure ($R^2 = 0.949$) raises ES₉₅ coverage to **95.9%** — a portable, deployable quoting-and-risk tool using only the asset's own options. Three supporting results frame these contributions: a fully arbitrage-free SSVI calibration database (2,633 daily surfaces, 95.1% success); a panel decomposition in which moneyness, maturity and day effects explain 86% of IV cross-sectional variation (96% with Surface-Cell FE) — SSVI supplies shape, not signal; and a single narrow predictability result, the wing-decay parameter γ , whose modest mean-reversion is fully captured by a plain MA(1). Every core econometric technique is exercised on the data, and the two extensions embed the standard toolbox in a genuine quantitative-finance application.

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